

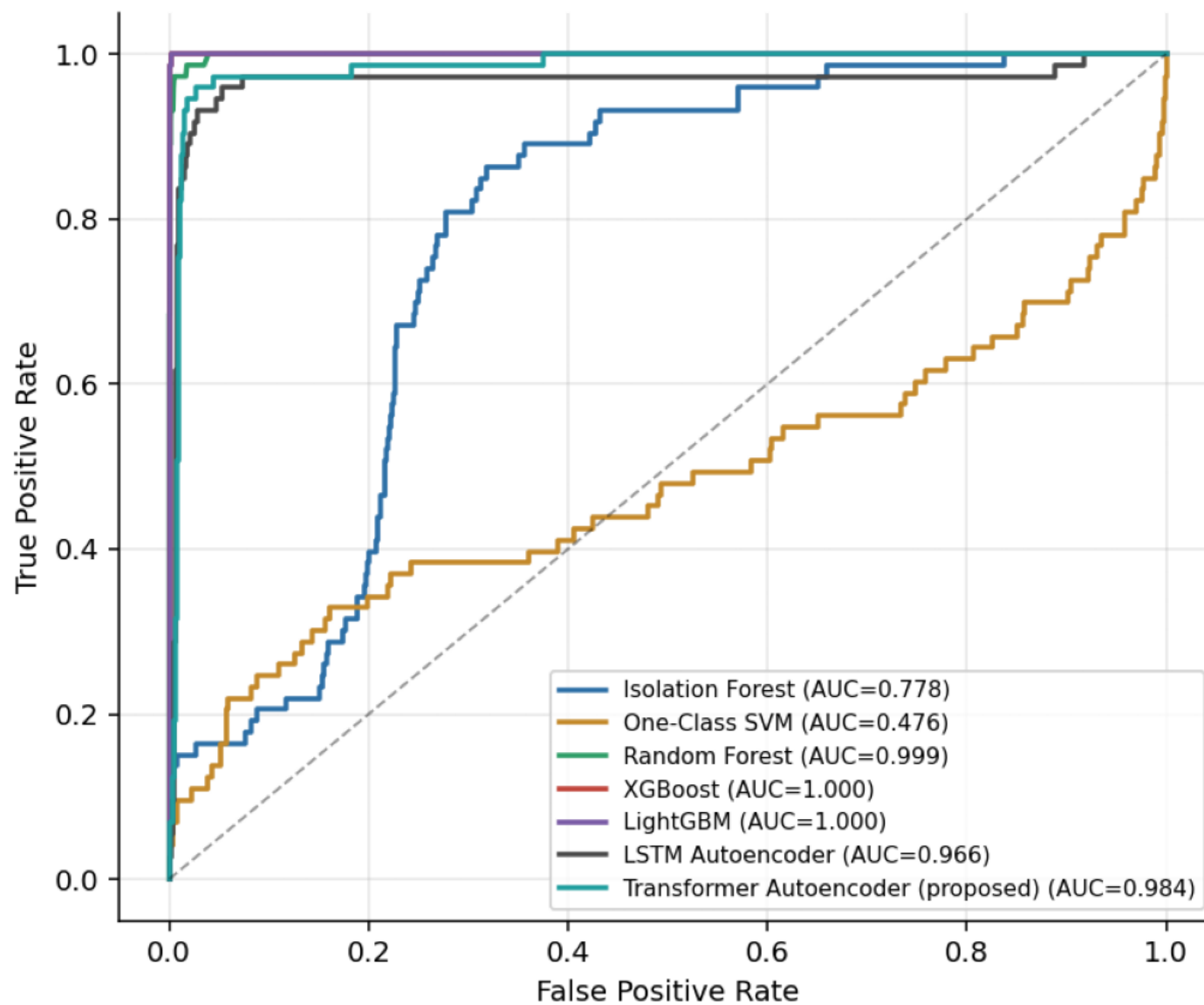
Smart Grid Cyberattack Detection — Benchmark

Leak-free protocol · held-out test metrics

Transformer Autoencoder (pro	model	precision	recall	f1	roc_auc	pr_auc	mcc
	XGBoost	1.000	0.973	0.986	1.000	0.999	0.986
	LightGBM	0.973	0.986	0.980	1.000	0.999	0.979
	Random Forest	0.962	0.685	0.800	0.999	0.961	0.809
	Transformer Autoencoder (pro	0.421	0.945	0.582	0.984	0.484	0.622
	LSTM Autoencoder	0.319	0.932	0.476	0.966	0.505	0.533
	One-Class SVM	0.057	0.219	0.090	0.476	0.068	0.079
	Isolation Forest	0.043	0.890	0.082	0.778	0.133	0.143

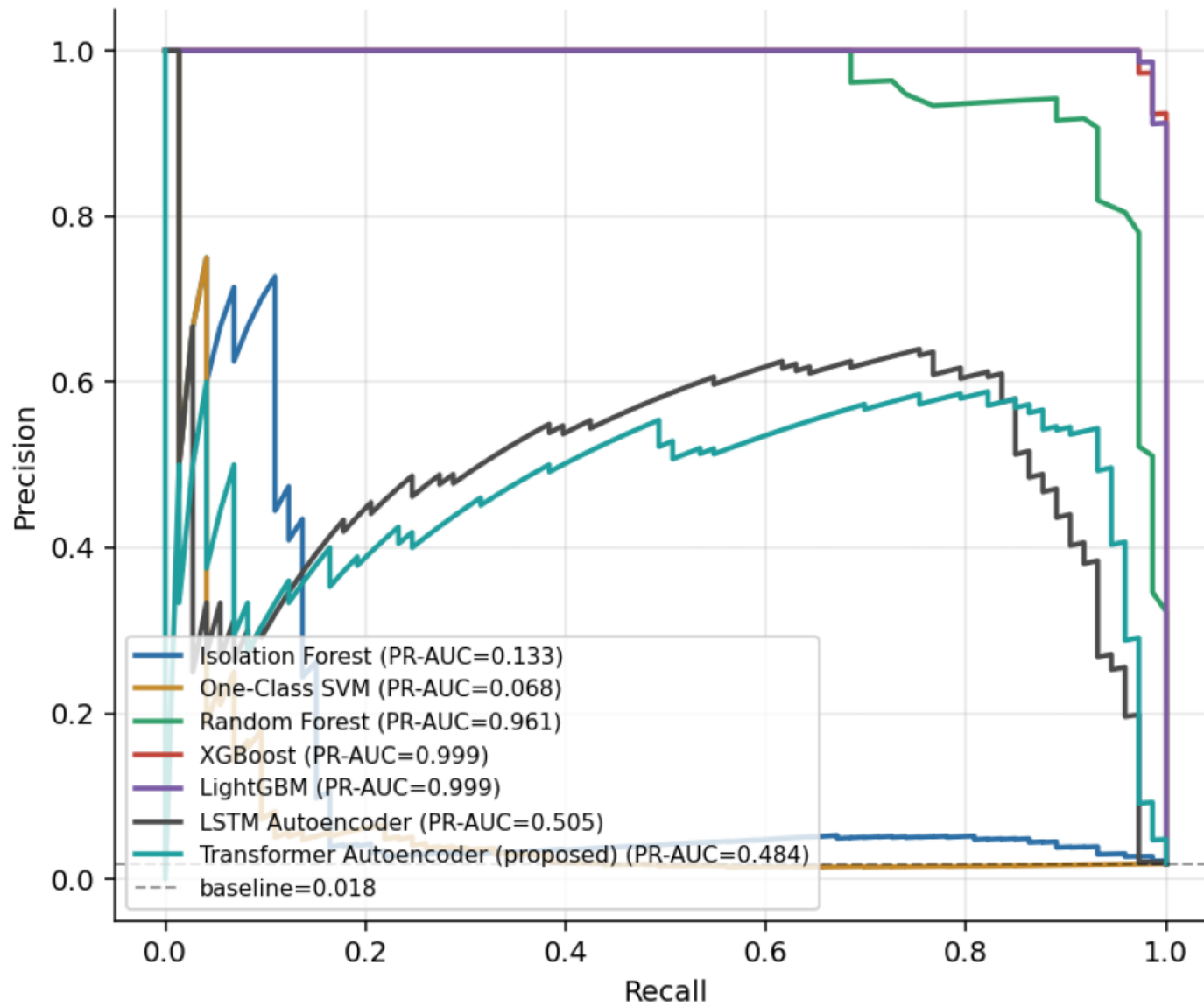
roc_overlay

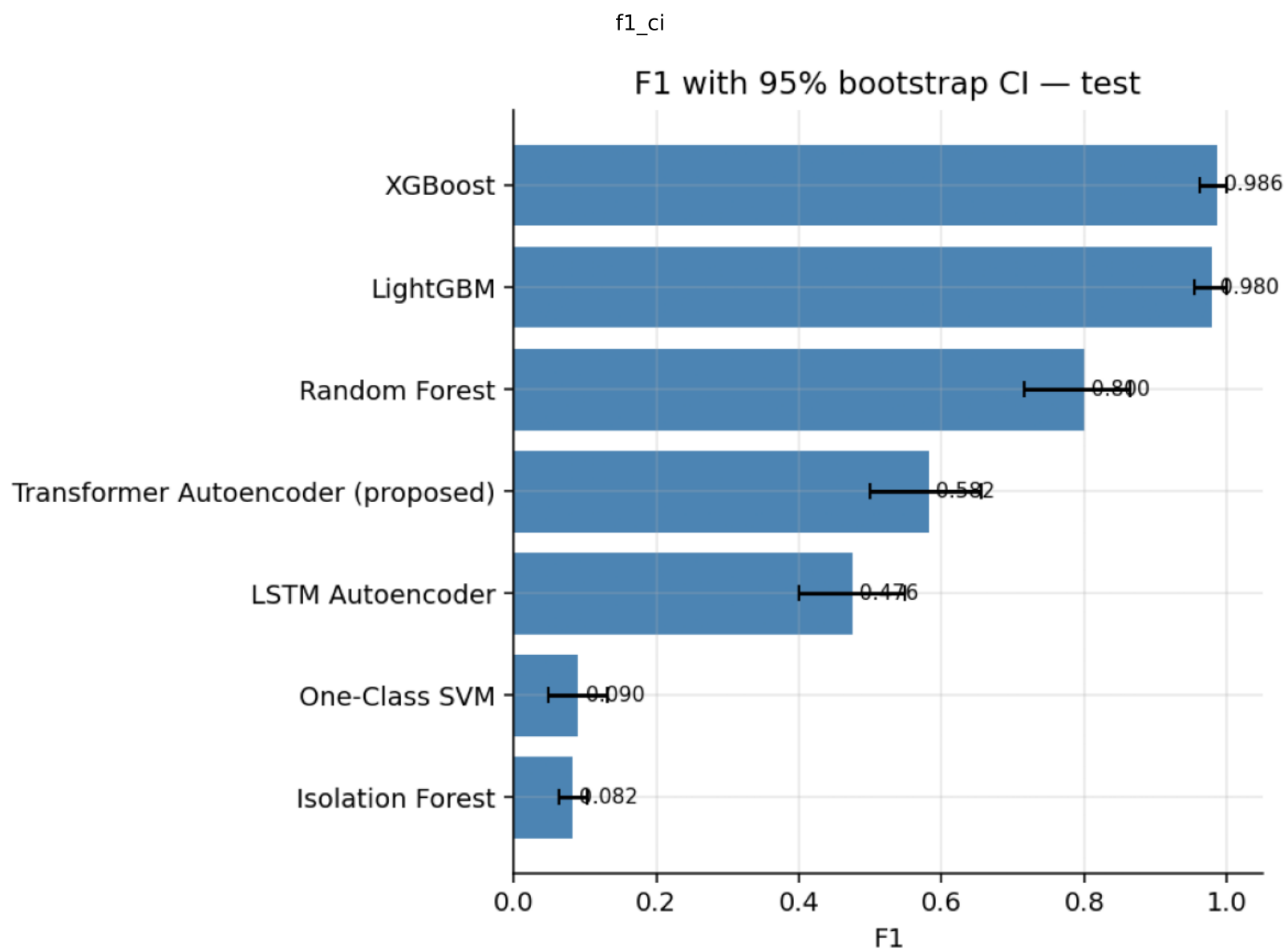
ROC — held-out test



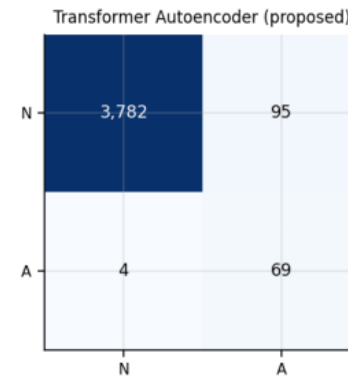
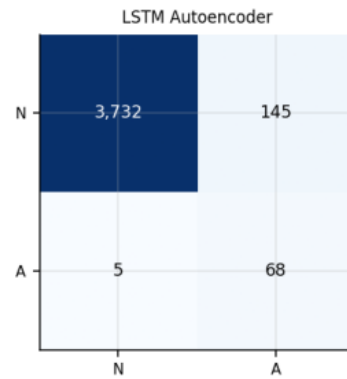
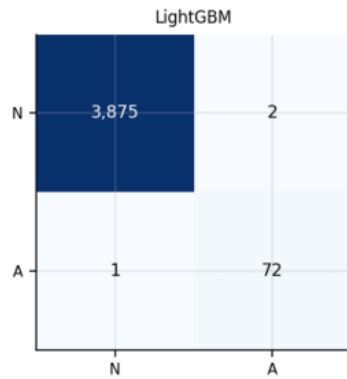
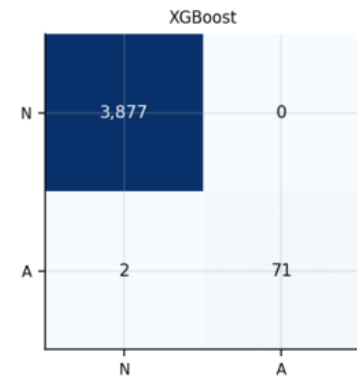
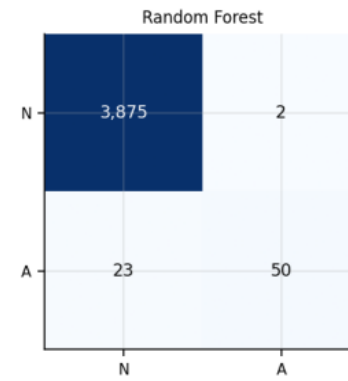
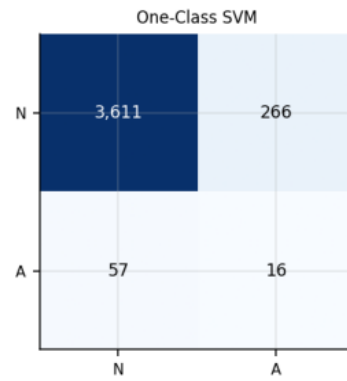
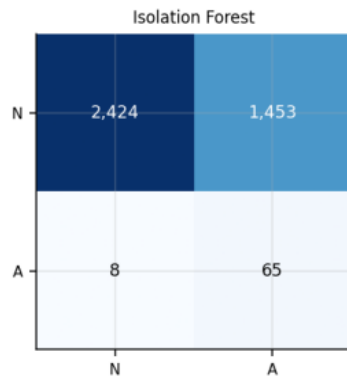
pr_overlay

Precision-Recall — held-out test

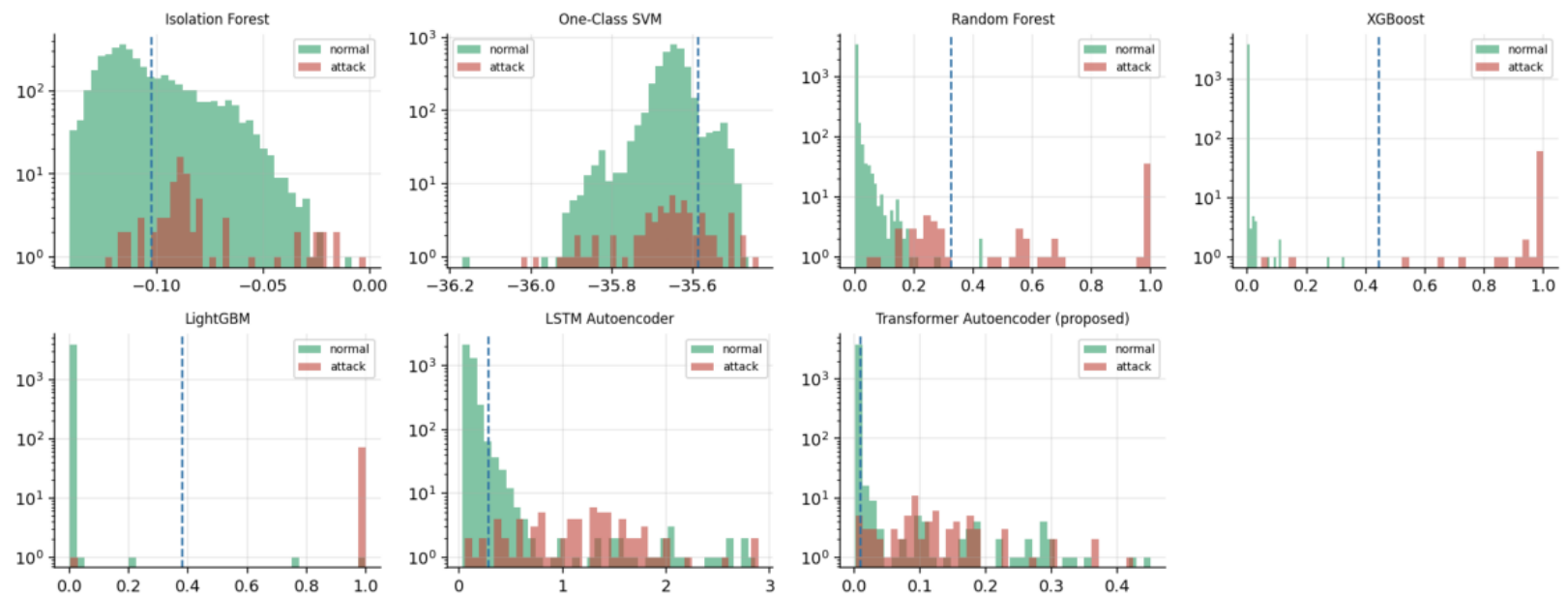




confusion_grid
Confusion matrices — held-out test

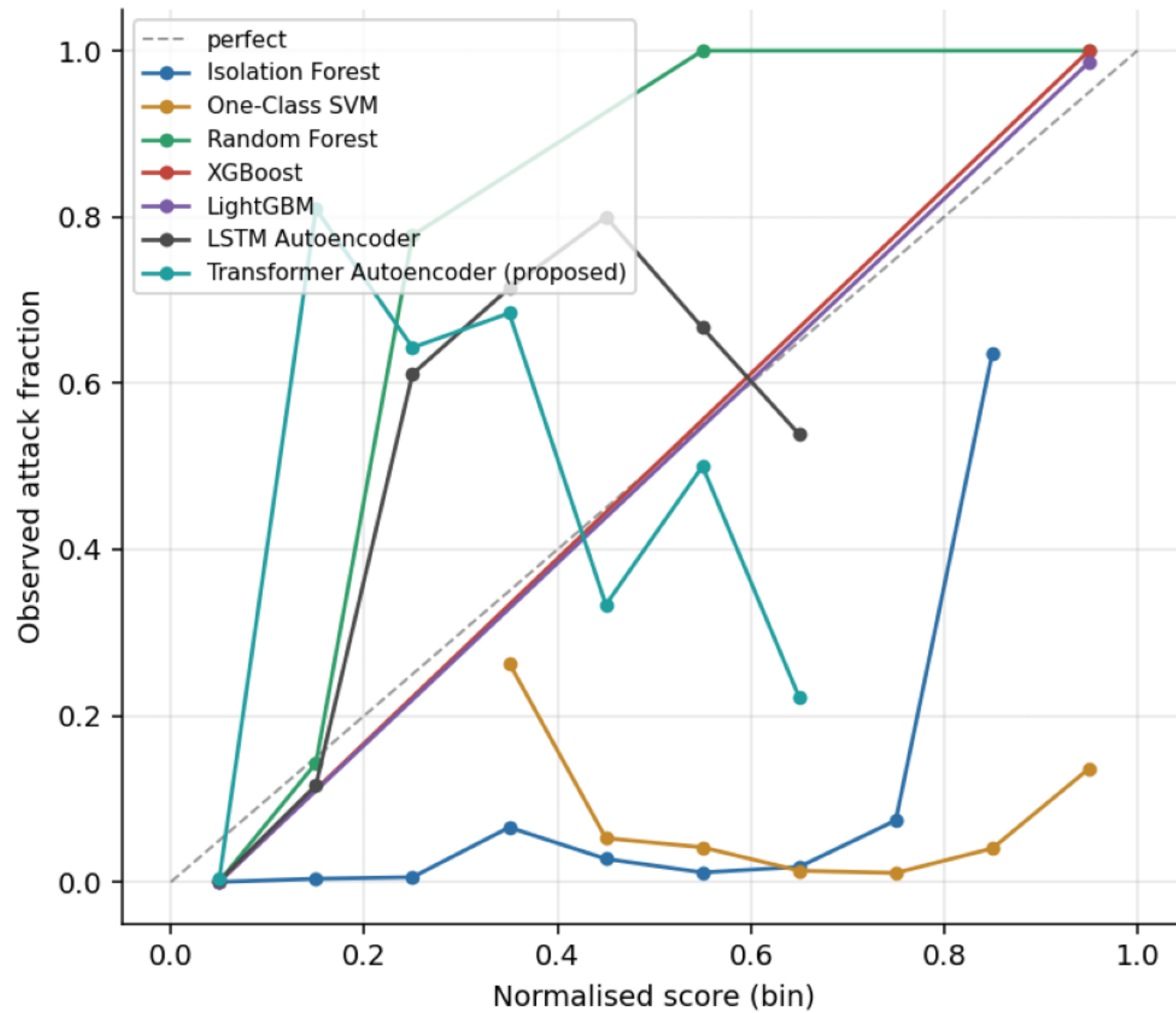


score_dists
Score distributions + threshold — test



calibration

Calibration — held-out test



feature_importance

